Q4 Help Doc: Kernel Build and Hooks

Target: Linux 6.8 on Ubuntu 24.04. **Take a VM snapshot first and keep your stock kernel in GRUB.**

1. Build the kernel

```
sudo apt install -y build-essential libncurses-dev bison flex libssl-dev \
                    libelf-dev dwarves zstd bc forkstat
wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.8.tar.xz
tar xf linux-6.8.tar.xz && cd linux-6.8

cp /boot/config-$(uname -r) .config
scripts/config --disable SYSTEM_TRUSTED_KEYS
scripts/config --disable SYSTEM_REVOCATION_KEYS
scripts/config --disable DEBUG_INFO
scripts/config --disable DEBUG_INFO_BTF
scripts/config --set-str LOCALVERSION "-forkguard"
scripts/config --enable SATA_AHCI        # disk driver must be built in
(=y),
scripts/config --enable VIRTIO_BLK       # otherwise boot drops to
(initramfs)
make olddefconfig

make -j$(nproc)
sudo make modules_install install
sudo update-grub && sudo reboot
```

Check with `uname -r` (must end in `-forkguard`). Finish your config before the first build; changing config later can trigger a full rebuild. Use 4 cores and 6 GB RAM or more.

If boot drops to `(initramfs)`, type `reboot` and pick your old kernel under *Advanced options* in GRUB.

2. Where to hook

File	Function	Purpose
`kernel/fork.c`	`kernel_clone()` (top)	Fork containment; return a negative errno to refuse the fork
`kernel/sched/core.c`	`sched_fork()`	Apply the user's tier as a nice value to new tasks
`kernel/Makefile`		Add `obj-y += fork_guard.o`

Put your logic in a new `kernel/fork_guard.c` with a header in `include/linux/`. The containment check must run before, and must not depend on, the tier logic.

3. Interfaces you will need

- **User and time:** `current_uid()`, `uid_eq()`, `GLOBAL_ROOT_UID`, `jiffies`, `msecs_to_jiffies()`, `time_after()`
- **Locking:** `spin_lock_irqsave()`. Never sleep while holding it.
- **Process group:** `task_pgrp()`, `kill_pgrp(pid, sig, 1)`, `get_pid()` / `put_pid()`
- **Scheduler:** `p->static_prio`, `NICE_TO_PRIO()`, `set_load_weight()`, `set_user_nice()`, `for_each_process()` with `rcu_read_lock()`
- **/proc:** `proc_create()`, `struct proc_ops`, `seq_file` (`single_open`, `seq_printf`)
- **sysctl:** `register_sysctl("kernel", table)`, `proc_douintvec_minmax` (6.8 needs an empty `{ }` sentinel at the end of the table)
- **Logging:** `pr_warn()`, `pr_alert()`

4. Points to think about

- `sched_fork()` only runs when a task is created. How will you affect a task that is already running?
- Why kill the process group and not one process? In what order should you send signals to a stopped group?
- CFS weights: nice -7 gives weight 4904, nice 0 gives 1024, nice +5 gives 335. Pick GOLD and BRONZE values that never starve anyone.

5. Userspace tool

`tier set <user> <gold|silver|bronze>`, `tier reset`, `tier show`. Convert the name to a uid with `getpwnam()` and write a command to your `/proc` control file.

6. Debugging

```
dmesg -w | grep fork_guard     # live kernel log
sudo forkstat                  # live process creation
ps -u alice -o pid,ni,pcpu,etime,comm
```

Test with a low fork threshold first so a bug does not wedge the VM.